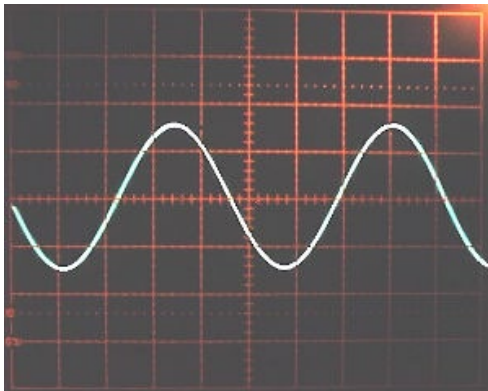


ECE 3317

Applied Electromagnetic Waves

Prof. David R. Jackson
Fall 2026



Notes 2

Complex Vectors

Adapted from notes by Prof. Stuart A. Long

Notation

Circuit quantities:

- $v(t)$ is a time-varying function.
- V is a phasor (complex number).

Note:

“Handscript SF” font is used for time-domain vector quantities.
(This font has been placed on Canvas for you.)

Field quantities:

- $\underline{\mathcal{E}}(t)$ is a time-varying vector function.
- $\underline{E}$ is a phasor vector (complex vector).
- $\mathcal{E}_x(t)$ is a time-varying component of a vector function.
- E_x is a phasor component of a vector function.

Appendix B in the Hayt & Buck book discusses units in some detail.

Complex Numbers

$$c = a + j b = |c| e^{j\phi}$$

↑
↑
↑
↑

Real part Imaginary part Magnitude Phase (always in radians)

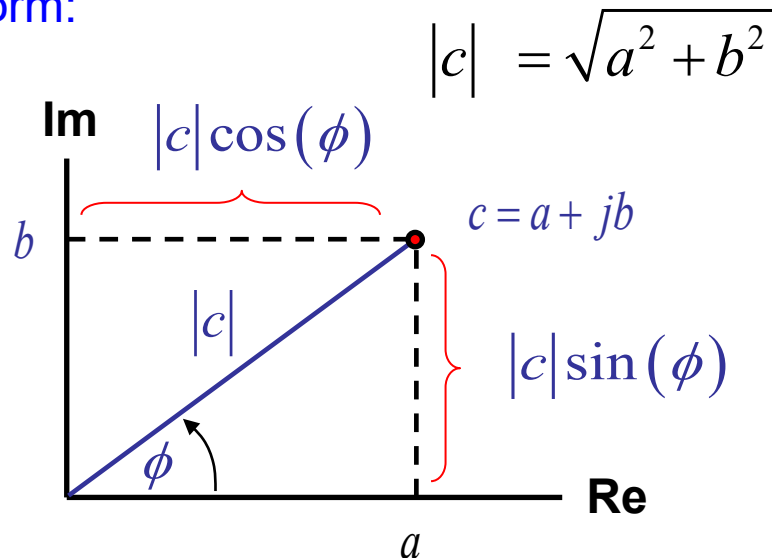
$j = \sqrt{-1}$

Note: a , b , ϕ are real.



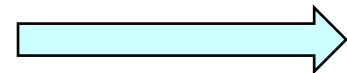
Euler's identity allows us to write the polar form.

Polar form:



Euler's identity:

$$e^{j\phi} = \cos \phi + j \sin \phi$$



$$c = |c| (\cos \phi + j \sin \phi)$$

$$a = |c| \cos \phi$$

$$b = |c| \sin \phi$$

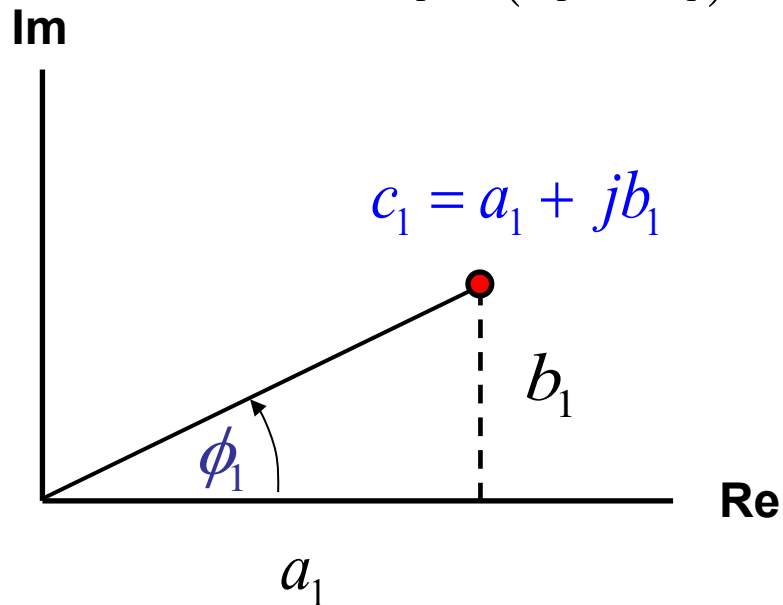
Note: The angle ϕ is measured positive going counterclockwise.

Complex Numbers (cont.)

Complex conjugate

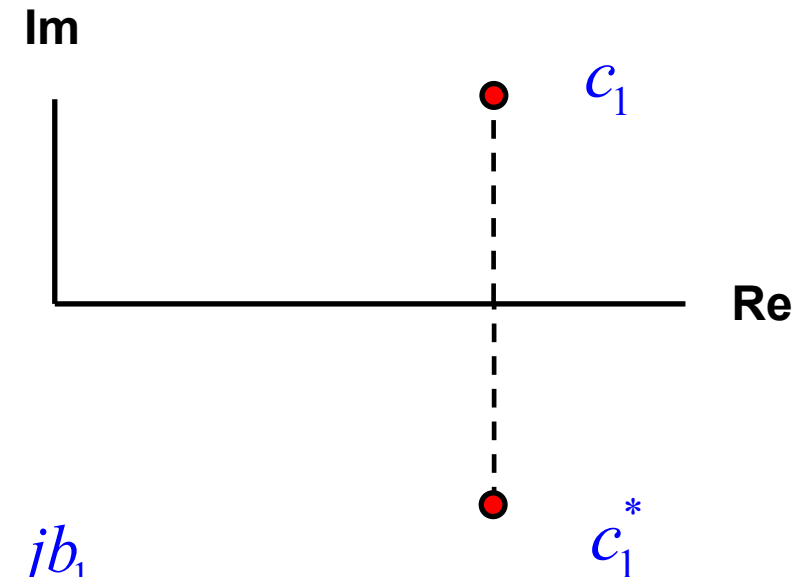
$$c_1 = (a_1 + jb_1) = |c_1| e^{j\phi_1}$$

$$c_1^* = (a_1 + jb_1)^* \equiv a_1 - jb_1$$



$$c_2 = c_1^* = a_2 + jb_2 = |c_2| e^{j\phi_2}$$

$$|c_2| = |c_1| \quad \phi_2 = -\phi_1$$



$$c_2 = a_2 + jb_2 = c_1^* = a_1 - jb_1$$

Complex Algebra

$$\begin{aligned}c_1 &= a_1 + jb_1 = |c_1| e^{j\phi_1} \\c_2 &= a_2 + jb_2 = |c_2| e^{j\phi_2}\end{aligned}$$

Addition

$$c_1 + c_2 = (a_1 + a_2) + j(b_1 + b_2)$$

Subtraction

$$c_1 - c_2 = (a_1 - a_2) + j(b_1 - b_2)$$

Complex Algebra (cont.)

$$\begin{aligned}c_1 &= a_1 + jb_1 = |c_1| e^{j\phi_1} \\c_2 &= a_2 + jb_2 = |c_2| e^{j\phi_2}\end{aligned}$$

Multiplication

$$(c_1)(c_2) = (a_1 + jb_1)(a_2 + jb_2) = (a_1a_2 - b_1b_2) + j(a_1b_2 + a_2b_1)$$

$$(c_1)(c_2) = |c_1||c_2|e^{j(\phi_1 + \phi_2)}$$

Division

$$\frac{c_1}{c_2} = \frac{a_1 + jb_1}{a_2 + jb_2} = \frac{a_1 + jb_1}{a_2 + jb_2} \left(\frac{a_2 - jb_2}{a_2 - jb_2} \right) = \frac{(a_1 + jb_1)(a_2 - jb_2)}{a_2^2 + b_2^2} = \frac{(a_1a_2 + jb_1b_2) + j(b_1a_2 - a_1b_2)}{a_2^2 + b_2^2}$$

$$\frac{c_1}{c_2} = \frac{|c_1|}{|c_2|} e^{j(\phi_1 - \phi_2)}$$

Square Root

Principal square root of a complex number (denoted by radical sign):

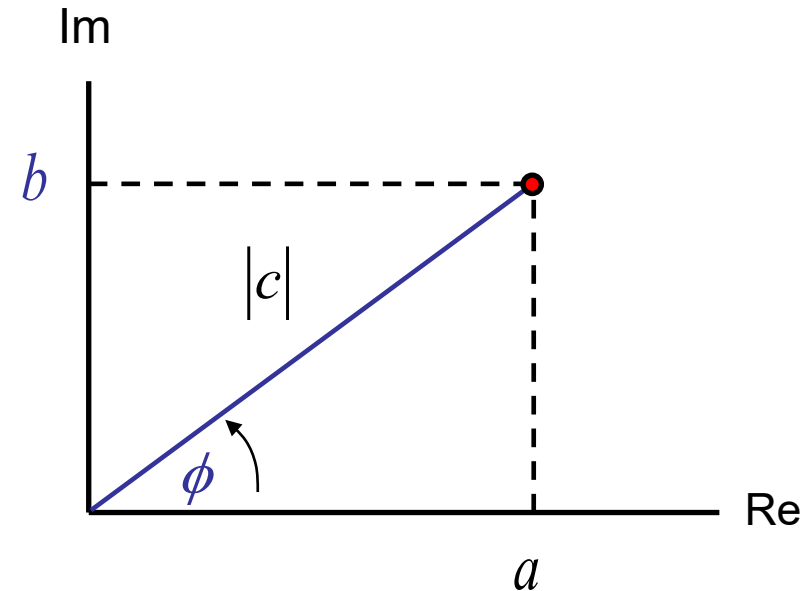
$$c = |c| e^{j\phi}$$

$$\sqrt{c} = \left(|c| e^{j\phi}\right)^{1/2} \quad -\pi < \phi \leq \pi \quad (\text{the principal branch})$$
$$\sqrt{|c|} e^{j(\phi/2)}$$

Note:

For a positive real number x , the principal square root is positive:

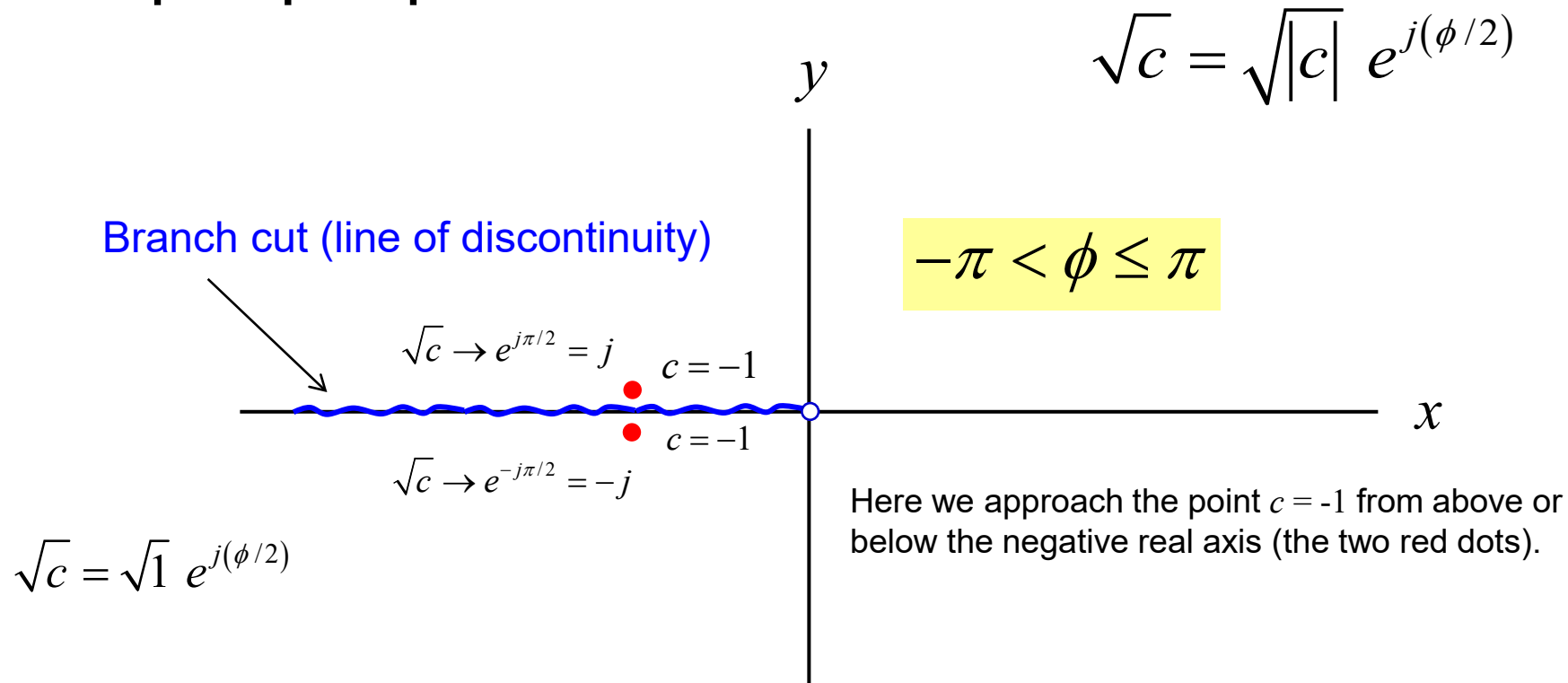
Example: $\sqrt{4} = \left(4e^{j(0)}\right)^{1/2} = 2e^{j(0)} = 2$



Square Root

$$c = |c| e^{j\phi}$$

Illustration of principal square root:

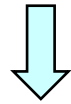


Note: Matlab uses the principal square root: $\sqrt{-1} = +j$

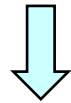
Square Root (cont.)

Property of principal square root:

$$-\pi < \phi \leq \pi$$



$$-\pi / 2 < \phi / 2 \leq \pi / 2$$



$$\operatorname{Re} \sqrt{c} \geq 0$$

$$\begin{aligned} c &= |c| e^{j\phi} \\ \sqrt{c} &= \sqrt{|c|} e^{j(\phi/2)} \\ -\pi < \phi &\leq \pi \end{aligned}$$

Examples :

$$\sqrt{4} = 2$$

$$\sqrt{j} = \sqrt{1e^{j\pi/2}} = \sqrt{1} e^{j\pi/4} = \frac{1+j}{\sqrt{2}}$$

Square Root (cont.)

$$c = |c| e^{j\phi}$$

General square root of a complex number (two values):

$$\begin{aligned} c^{1/2} &= (|c| e^{j\phi})^{1/2} \\ &= (|c| e^{j(\phi_p + 2\pi n)})^{1/2} \quad \left(-\pi < \phi_p \leq \pi, \quad n \text{ is any integer} \right) \\ &= \sqrt{|c|} e^{j(\phi_p/2 + 2\pi n/2)} \\ &= \left(\sqrt{|c|} e^{j\phi_p/2} \right) e^{jn\pi} \\ &= \pm \sqrt{|c|} e^{j\phi_p/2} \quad (+ \text{ for } n \text{ even, } - \text{ for } n \text{ odd}) \\ &= \pm \sqrt{c} \end{aligned}$$

Denotes the angle of the principal branch
↓

Examples:

$$4^{1/2} = \pm 2$$

$$j^{1/2} = \pm \left(\frac{1+j}{\sqrt{2}} \right)$$

The general square root has two possible values.

Time-Harmonic Quantities

$$v(t) = A \cos(\omega t + \phi)$$

Amplitude

Angular
Frequency

Phase

Note: A , ω , ϕ are real.

$$f = \frac{\omega}{2\pi}$$

Frequency [Hz]

$$T = \frac{1}{f}$$

Period [s]

From Euler's identity:

$$v(t) = \operatorname{Re}\left\{A e^{j(\phi + \omega t)}\right\} = \operatorname{Re}\left\{\left(A e^{j\phi}\right) e^{j\omega t}\right\}$$

Define the phasor : $V \equiv A e^{j\phi}$

We then have:

$$v(t) = \operatorname{Re}\left\{V e^{j\omega t}\right\}$$

Note:

The phase angle of the sinusoidal wave is the same as the phase angle of the complex number (phasor). The amplitude of the sinusoidal wave is the same as the magnitude of the complex number (phasor).

Time-Harmonic Quantities (cont.)

$$v(t) = A \cos(\omega t + \phi)$$

$$\left\{ \begin{array}{ll} V \equiv A e^{j\phi} & \text{going from time domain to phasor domain} \\ v(t) = \operatorname{Re}\{V e^{j\omega t}\} & \text{going from phasor domain to time domain} \end{array} \right.$$

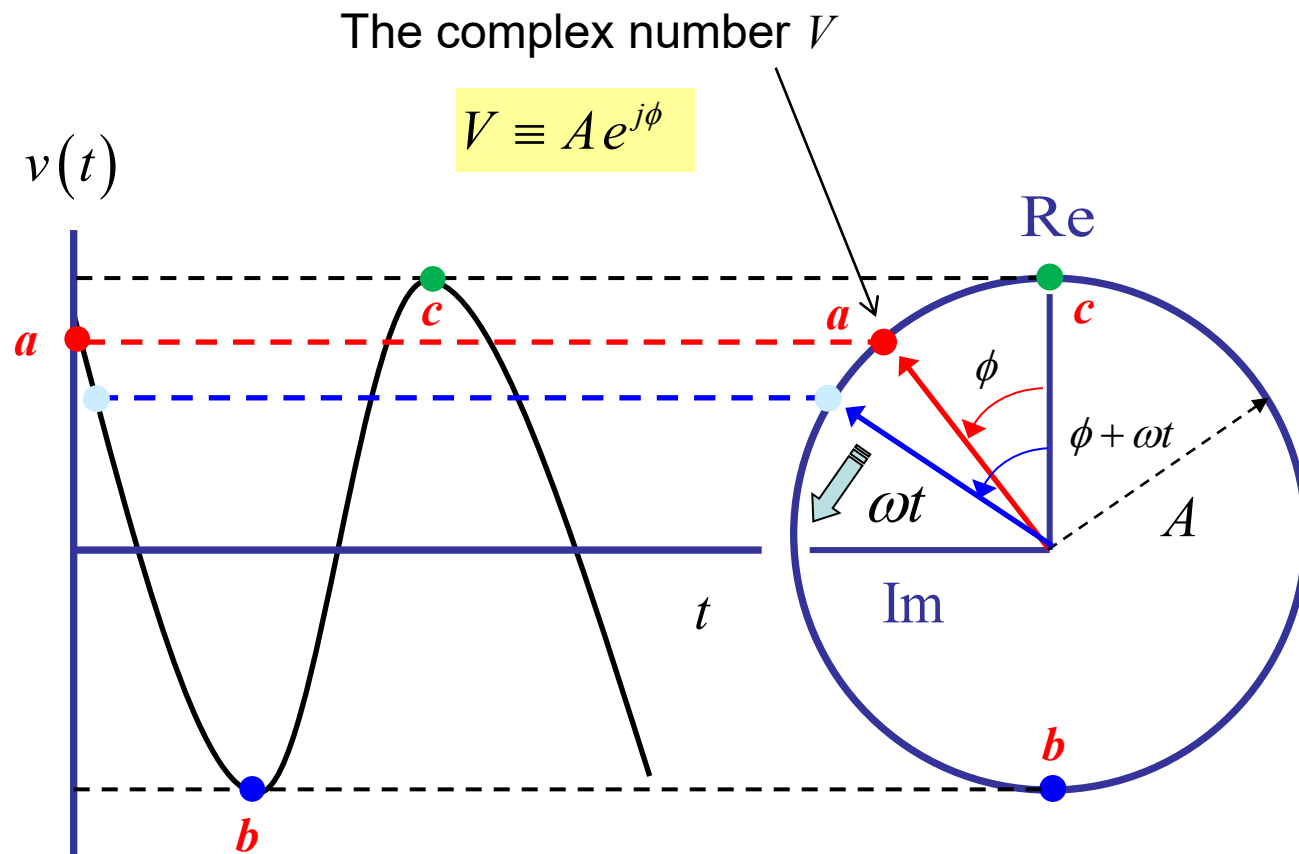
Time-domain $\Leftrightarrow$ Phasor domain

$$v(t) \Leftrightarrow V$$

Time-Harmonic Quantities (cont.)

$$v(t) = A \cos(\omega t + \phi)$$

$$\begin{aligned} v(t) &= \operatorname{Re}\{V e^{j\omega t}\} \\ &= \operatorname{Re}\{A e^{j\phi} e^{j\omega t}\} \end{aligned}$$



Time-Harmonic Quantities (cont.)

$$v(t) \Leftrightarrow V$$

Note:

$$u(t) + v(t) \Leftrightarrow U + V \quad \text{This assumes that the two sinusoidal signals are at the same frequency.}$$

$$\frac{\partial}{\partial t} v(t) \Leftrightarrow j\omega V \quad \text{There are no time derivatives in the phasor domain!}$$

However:

$$u(t)v(t) \not\Leftrightarrow UV \quad \left(\text{i.e., } u(t)v(t) \neq \text{Re}(UVe^{j\omega t}) \right)$$

All phasors are complex numbers, but not all complex numbers are phasors!

Complex Vectors

Vectors in the Phasor Domain

$$\begin{aligned}\underline{\mathcal{E}}(t) &= \underline{\hat{x}}\underline{\mathcal{E}}_x(t) + \underline{\hat{y}}\underline{\mathcal{E}}_y(t) + \underline{\hat{z}}\underline{\mathcal{E}}_z(t) \quad (\text{assumed to be sinusoidal}) \\ &= \underline{\hat{x}}A_x \cos(\omega t + \phi_x) + \underline{\hat{y}}A_y \cos(\omega t + \phi_y) + \underline{\hat{z}}A_z \cos(\omega t + \phi_z)\end{aligned}$$

Convert to phasor domain:

$$\begin{aligned}\underline{\mathcal{E}}(t) &= \text{Re}\left(\underline{\hat{x}}A_x e^{j\phi_x} e^{j\omega t}\right) + \text{Re}\left(\underline{\hat{y}}A_y e^{j\phi_y} e^{j\omega t}\right) + \text{Re}\left(\underline{\hat{z}}A_z e^{j\phi_z} e^{j\omega t}\right) \\ &= \text{Re}\left\{\left(\underline{\hat{x}}A_x e^{j\phi_x} + \underline{\hat{y}}A_y e^{j\phi_y} + \underline{\hat{z}}A_z e^{j\phi_z}\right)e^{j\omega t}\right\} \\ &= \text{Re}\left\{\left(\underline{\hat{x}}E_x + \underline{\hat{y}}E_y + \underline{\hat{z}}E_z\right)e^{j\omega t}\right\} \\ &= \text{Re}\left\{\underline{E} e^{j\omega t}\right\}\end{aligned}$$

where

$$\underline{E} \equiv \underline{\hat{x}}E_x + \underline{\hat{y}}E_y + \underline{\hat{z}}E_z \quad \text{Complex vector!}$$

Question:

Why does the frequency have to be the same for all components?

Complex Vectors (cont.)

$$\underline{\mathcal{E}}(t) = \hat{x}\underline{\mathcal{E}}_x(t) + \hat{y}\underline{\mathcal{E}}_y(t) + \hat{z}\underline{\mathcal{E}}_z(t) \quad (\text{sinusoidal})$$

We have proven:

$$\underline{\mathcal{E}}(t) = \text{Re} \left\{ \underline{E} e^{j\omega t} \right\}$$

We thus work with phasor vectors the same way as we do with phasor scalars!

where

$$\underline{E} \equiv \hat{x}E_x + \hat{y}E_y + \hat{z}E_z \quad \text{Complex vector!}$$

Notation:

$$\underline{\mathcal{E}}(t) \Leftrightarrow \underline{E}$$

$$\mathcal{E}_x(t) \Leftrightarrow E_x$$

$$\mathcal{E}_y(t) \Leftrightarrow E_y$$

$$\mathcal{E}_z(t) \Leftrightarrow E_z$$

Example 1.15 (Shen & Kong)

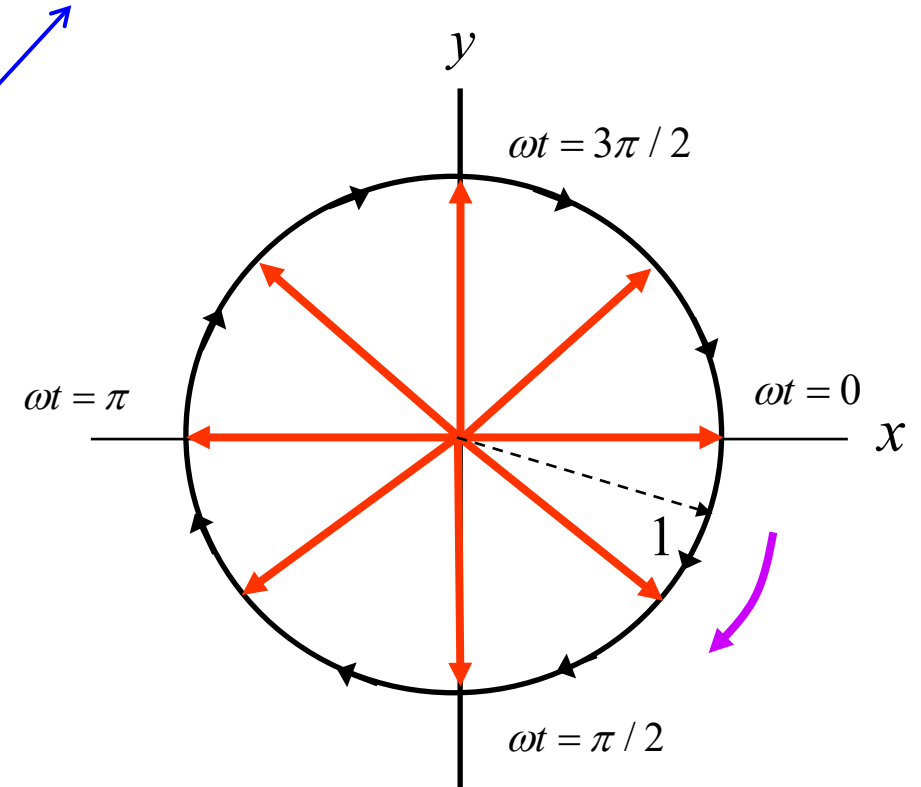
Assume $\underline{A} = \underline{\hat{x}} + j\underline{\hat{y}}$

Note: $|\underline{\mathcal{A}}(t)| = \sqrt{\mathcal{A}_x^2(t) + \mathcal{A}_y^2(t)} = \sqrt{\cos^2 \omega t + \sin^2 \omega t} = 1$

Find the corresponding time-domain vector.

$$\begin{aligned}\underline{\mathcal{A}}(t) &= \text{Re}\{\underline{A}e^{j\omega t}\} \\ &= \text{Re}\{(\underline{\hat{x}} + j\underline{\hat{y}})e^{j\omega t}\} \\ &= \text{Re}\{(\underline{\hat{x}} + j\underline{\hat{y}})(\cos \omega t + j \sin \omega t)\} \\ &= \underline{\hat{x}} \cos \omega t - \underline{\hat{y}} \sin \omega t\end{aligned}$$

$$\underline{\mathcal{A}}(t) = \underline{\hat{x}} \cos \omega t - \underline{\hat{y}} \sin \omega t$$



The vector rotates (clockwise) with time!

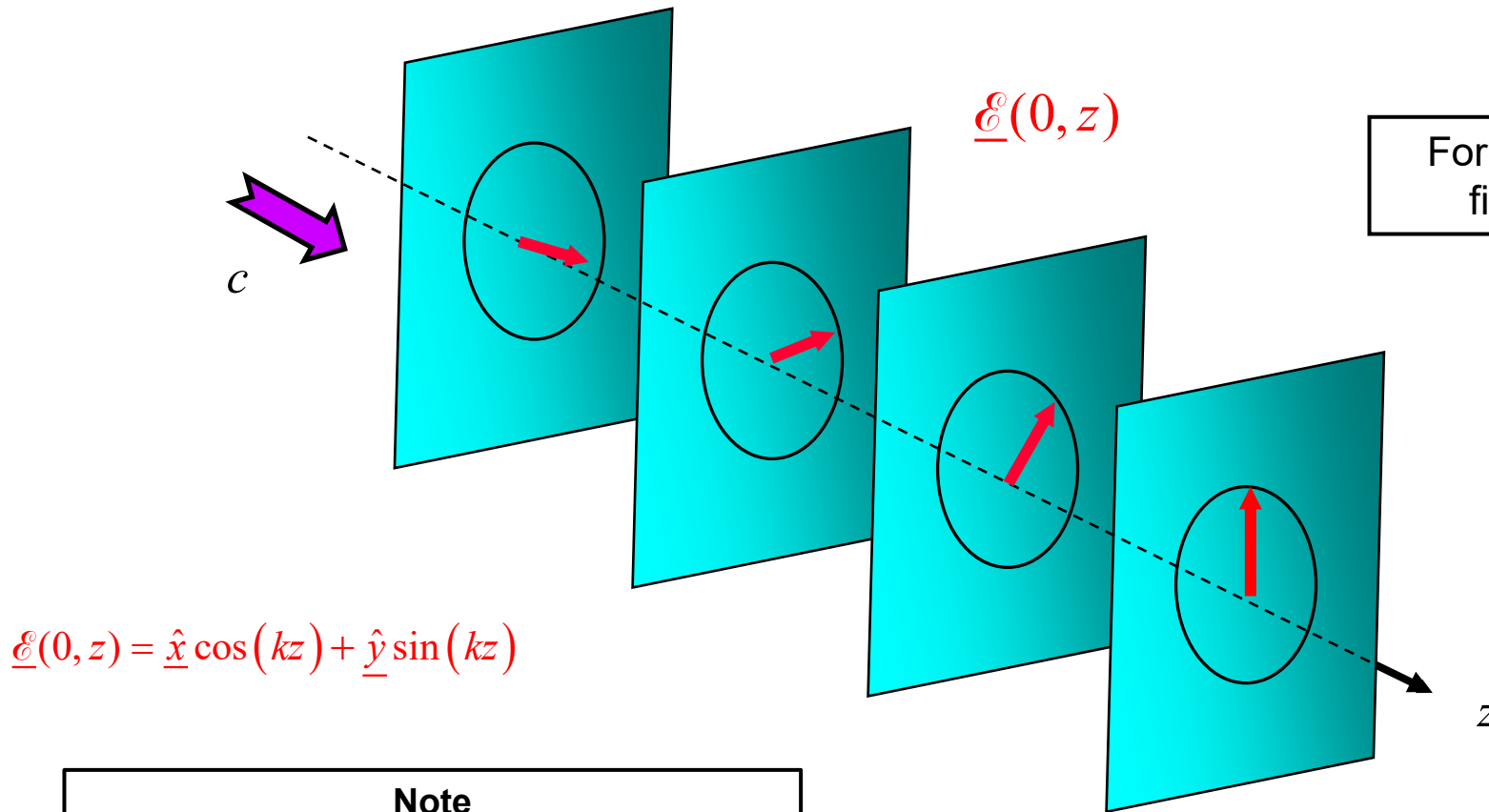
Note: The vector rotates counterclockwise if the minus sign is changed to a plus sign.

Example 1.15 (cont.)

Practical application:

A circular-polarized plane wave (discussed later)

$$\underline{\mathcal{E}}(t, z) = \underline{\hat{x}} \cos(\omega t - kz) - \underline{\hat{y}} \sin(\omega t - kz)$$



$$\underline{\mathcal{E}}(0, z) = \underline{\hat{x}} \cos(kz) + \underline{\hat{y}} \sin(kz)$$

For a fixed value of position z , the electric field vector rotates clockwise in time.

$$\underline{\mathcal{E}}(t, 0) = \underline{\hat{x}} \cos(\omega t) - \underline{\hat{y}} \sin(\omega t)$$

Note
For a fixed value of time (“snapshot”), the field rotates counterclockwise in space.

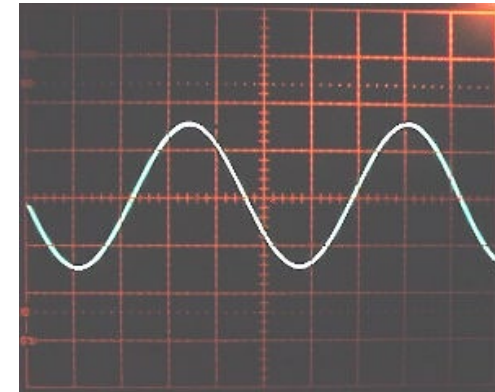
This is a “snapshot” of the electric field at $t = 0$.

Time Average of Time-Harmonic Quantities

$$v(t) = A \cos(\omega t + \phi)$$

$$\langle v(t) \rangle \equiv \frac{1}{T} \int_0^T A \cos(\omega t + \phi) dt = 0$$

$$\left(T = \frac{1}{f} = \frac{2\pi}{\omega} \right)$$



Note: $\cos^2 x = [1 + \cos(2x)] / 2$

$$\langle v^2(t) \rangle = \frac{1}{T} \int_0^T A^2 \cos^2(\omega t + \phi) dt$$

$$\langle v^2(t) \rangle = \frac{A^2}{T} \int_0^T \left\{ \frac{1 + \cos[2(\omega t + \phi)]}{2} \right\} dt$$

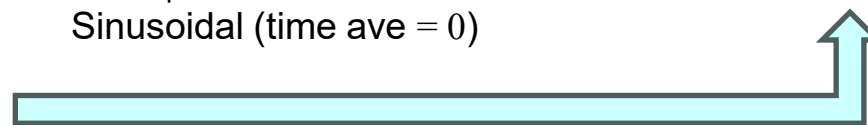
$$\langle v^2(t) \rangle = \frac{A^2}{T} \left(\frac{T}{2} \right) = \frac{A^2}{2}$$

Sinusoidal (time ave = 0)

Hence, we have:

$$\langle v^2(t) \rangle = \frac{A^2}{2}$$

(Note: The average value of $\cos^2$ is 1/2.)



Time Average of Time-Harmonic Quantities (cont.)

Next, consider the time average of a **product** of sinusoids:

$$\langle v(t)i(t) \rangle = \frac{1}{T} \int_0^T [A \cos(\omega t + \alpha)] [B \cos(\omega t + \beta)] dt$$

Note: $\cos x \cos y = [\cos(x - y) + \cos(x + y)] / 2$

$$\begin{aligned} \langle v(t)i(t) \rangle &= \frac{1}{T} AB \int_0^T \cos(\omega t + \alpha) \cos(\omega t + \beta) dt \\ &= \frac{AB}{T} \int_0^T \left[\frac{\cos(\alpha - \beta) + \cos(2\omega t + \alpha + \beta)}{2} \right] dt \\ &= \frac{AB}{T} \left[T \frac{\cos(\alpha - \beta)}{2} \right] \\ &= AB \left[\frac{\cos(\alpha - \beta)}{2} \right] \end{aligned}$$

↑
Sinusoidal (time ave = 0)

Time Average of Time-Harmonic Quantities (cont.)

Next, consider

$$\begin{aligned} VI^* &= \left(A e^{j\alpha} \right) \left(B e^{j\beta} \right)^* \\ &= \left(A e^{j\alpha} \right) \left(B e^{-j\beta} \right) \\ &= AB e^{j(\alpha - \beta)} \end{aligned}$$

Hence,

$$\operatorname{Re}(VI^*) = AB \cos(\alpha - \beta)$$

Recall that

$$\langle v(t)i(t) \rangle = AB \left[\frac{\cos(\alpha - \beta)}{2} \right] \quad (\text{from previous slide})$$

Hence

$$\langle v(t)i(t) \rangle = \frac{1}{2} \operatorname{Re}(VI^*)$$

Question:

Can we put the conjugate on the V instead of the I ?

Time Average of Time-Harmonic Quantities (cont.)

The results directly extend to **vectors** that vary sinusoidally in time.

Consider: $\underline{\mathcal{D}}(t) \cdot \underline{\mathcal{E}}(t) = (\mathcal{D}_x \mathcal{E}_x + \mathcal{D}_y \mathcal{E}_y + \mathcal{D}_z \mathcal{E}_z)$

$$\mathcal{D}_{x,y,z}(t) = \text{Re} \left[D_{x,y,z} e^{j\omega t} \right] \text{ etc.}$$

$$\begin{aligned} \langle \underline{\mathcal{D}} \cdot \underline{\mathcal{E}} \rangle &= \langle \mathcal{D}_x \mathcal{E}_x + \mathcal{D}_y \mathcal{E}_y + \mathcal{D}_z \mathcal{E}_z \rangle = \langle \mathcal{D}_x \mathcal{E}_x \rangle + \langle \mathcal{D}_y \mathcal{E}_y \rangle + \langle \mathcal{D}_z \mathcal{E}_z \rangle \\ &= \frac{1}{2} \text{Re} (D_x E_x^*) + \frac{1}{2} \text{Re} (D_y E_y^*) + \frac{1}{2} \text{Re} (D_z E_z^*) \\ &= \frac{1}{2} \text{Re} (D_x E_x^* + D_y E_y^* + D_z E_z^*) \end{aligned}$$

Hence $\langle \underline{\mathcal{D}}(t) \cdot \underline{\mathcal{E}}(t) \rangle = \frac{1}{2} \text{Re} (\underline{D} \cdot \underline{E}^*)$

Time Average of Time-Harmonic Quantities (cont.)

The result holds for both **dot** product and **cross** products.

$$\langle \underline{\mathcal{D}}(t) \cdot \underline{\mathcal{E}}(t) \rangle = \frac{1}{2} \operatorname{Re}(\underline{D} \cdot \underline{E}^*)$$

$$\langle \underline{\mathcal{E}}(t) \times \underline{\mathcal{H}}(t) \rangle = \frac{1}{2} \operatorname{Re}(\underline{E} \times \underline{H}^*)$$

where

$$\underline{\mathcal{D}}(x, y, z, t) = \operatorname{Re}[\underline{D}(x, y, z)e^{j\omega t}] \text{ etc.}$$

Time Average of Time-Harmonic Quantities (cont.)

To illustrate, consider the time-average stored electric energy density [J/m³] for a sinusoidal electric field.

$$U_E(t) = \frac{1}{2} \underline{\mathcal{D}}(t) \cdot \underline{\mathcal{E}}(t) \quad (\text{from ECE 3318})$$

$$\begin{aligned} \langle U_E(t) \rangle &= \frac{1}{2} \langle \underline{\mathcal{D}}(t) \cdot \underline{\mathcal{E}}(t) \rangle \\ &= \frac{1}{2} \left[\frac{1}{2} \text{Re}(\underline{D} \cdot \underline{E}^*) \right] \end{aligned}$$

$$\langle U_E(t) \rangle = \frac{1}{4} \text{Re}(\underline{D} \cdot \underline{E}^*)$$

“Stored electric energy density”

Time Average of Time-Harmonic Quantities (cont.)

Similarly,

$$U_H(t) = \frac{1}{2} \underline{\mathcal{B}}(t) \cdot \underline{\mathcal{H}}(t) \quad \text{“Stored magnetic energy density”}$$

$$\langle U_H(t) \rangle = \frac{1}{4} \operatorname{Re}(\underline{B} \cdot \underline{H}^*)$$

$$\underline{\mathcal{S}}(t) \equiv \underline{\mathcal{E}}(t) \times \underline{\mathcal{H}}(t) \quad \text{“Poynting vector”}$$

$$\langle \underline{\mathcal{S}}(t) \rangle = \frac{1}{2} \operatorname{Re}(\underline{E} \times \underline{H}^*)$$